

BUCK CONVERTER

Project Name: Design of a DC Voltage Buck Converter

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Project Overview

Introduction:

This report presents the complete design, modeling, implementation, and evaluation of a voltage-mode buck (step-down) DC–DC converter intended for power management integrated circuit (PMIC) applications. The work is developed from first principles and is written for an audience without prior exposure to earlier coursework or intermediate project steps. Both theoretical foundations and simulation-based verification are used to demonstrate stable operation, efficient switching, and robust closed-loop control.

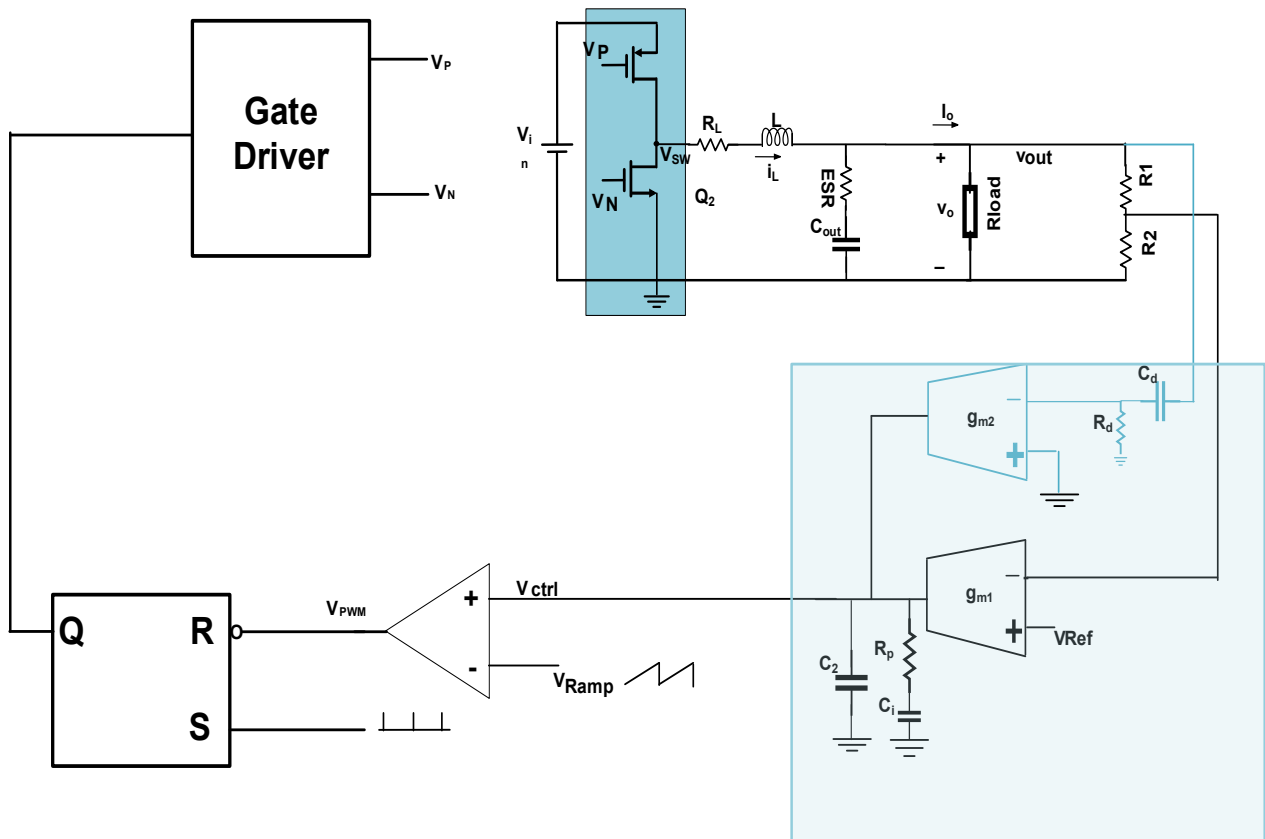


Fig. 1. Buck Topology for the design

The converter design integrates three major aspects: (i) power-stage modeling and small-signal analysis, (ii) control-loop compensation for stability and transient performance, and (iii) gate-driver and switching analysis for efficiency optimization. Supporting theory is drawn from established power electronics literature and MOSFET gate-driver **fundamentals**.

Problem Statement:

A buck converter used in modern PMICs must meet several competing requirements: accurate voltage regulation, closed-loop stability across load conditions, high efficiency, and reliable switching behavior at high frequency. Poor compensation can lead to instability or excessive ringing, while improper gate driving can cause hard switching, shoot-through, and large switching losses.

The problem addressed in this work is the **systematic design of a voltage-mode buck converter** that achieves stable closed-loop operation with controlled phase margin, minimized switching losses through soft switching, and high efficiency over a wide load range.

Requirements:

The converter is designed to meet the following specifications:

- Input voltage (V_{in}): 3.3 V
- Output voltage (V_{out}): 1.5 V
- Switching frequency (F_{sw}): 1.5 MHz
- Load current range: 0.1 A – 1 A
- Inductor current ripple: $\approx 20\%$ of maximum load current
- Control bandwidth target: $F_{sw} / 10$
- Phase margin targets: near 0° (marginal stability) and $\sim 60^\circ$ (robust stability)

Design and Implementation

Gate Driver and Switching:

The buck converter employs synchronous MOSFET switching using a high-side PMOS and a low-side NMOS device. A CMOS inverter-based gate driver is designed to provide sufficient drive strength while controlling the timing between switch transitions. Proper dead time is intentionally inserted between the turn-off of one switch and the turn-on of the other to prevent shoot-through and to enable soft switching.

MOSFET sizing is selected to balance conduction loss and gate capacitance. Larger devices reduce on-resistance but increase gate charge, placing greater demand on the driver. Dead time is generated using inverter delay chains, allowing the switch node voltage to settle before the complementary device turns on. This behavior directly reduces overlap between voltage and current during switching transitions.

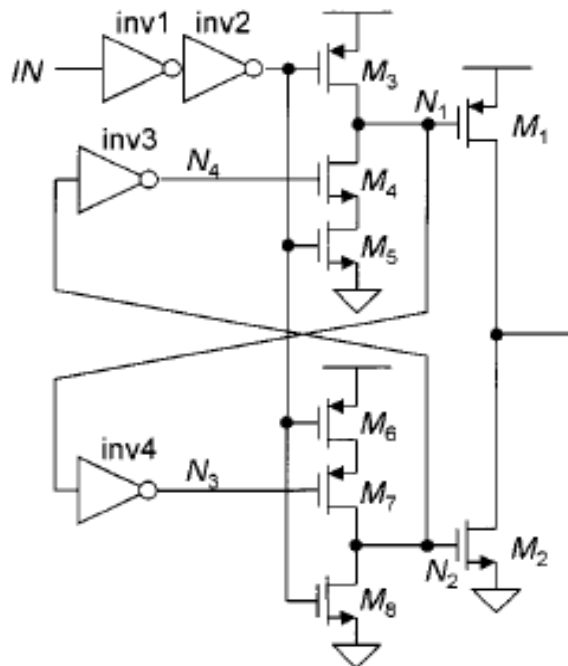


Fig. 2. Gate Driver Circuit

Compensator Stage:

Closed-loop regulation is achieved using a voltage-mode control architecture with an error amplifier and a Type-III compensation network. The compensation network introduces:

- An integrator to eliminate steady-state error,
- Zeros to counteract the LC double pole of the power stage,
- High-frequency poles to attenuate switching noise.

Two compensation designs are evaluated to highlight stability trade-offs:

- A low phase-margin design (near 0°) illustrating marginal stability and ringing,
- A higher phase-margin design ($\sim 60^\circ$) demonstrating well-damped transient behavior.

Pole and zero placement follows classical control techniques, including K-factor-based compensation methods.

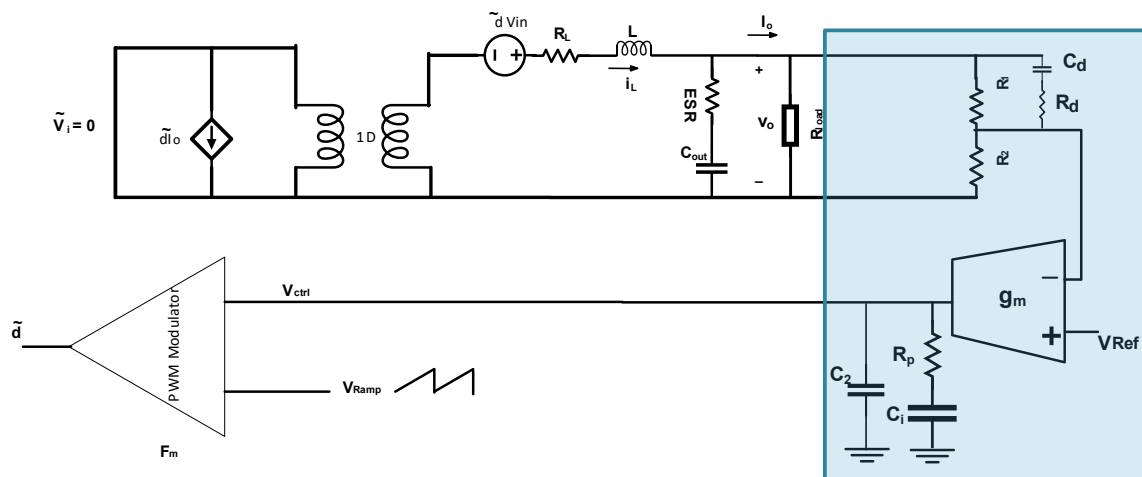


Fig. 3. Voltage Mode Buck Converter with Compensation Network – 1 (Low Phase Margin)

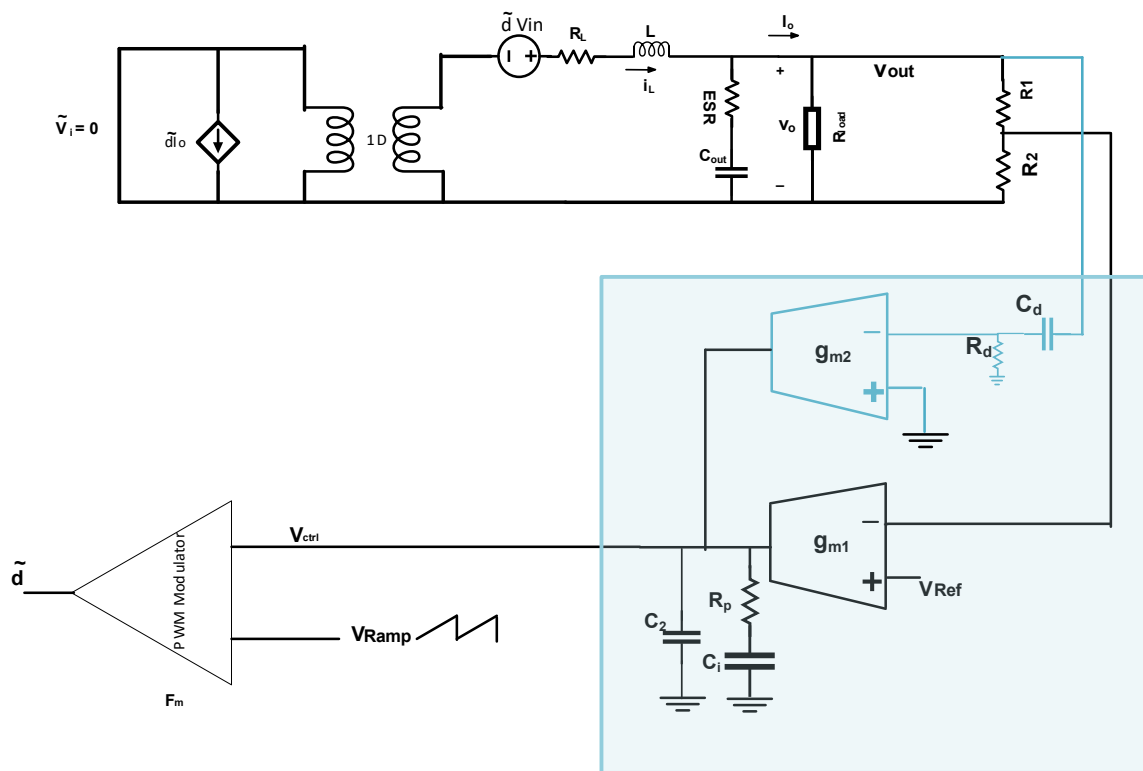


Fig. 4. Voltage Mode Buck Converter with Compensation Network –2 (High Phase Margin)

Digital Components:

Digital circuitry is used to generate the PWM control signal by comparing the error amplifier output with a periodic ramp signal. Additional digital inverter chains are used to generate dead time for the gate driver. These blocks illustrate the interaction between analog control loops and digital timing circuits commonly found in mixed-signal PMIC designs.

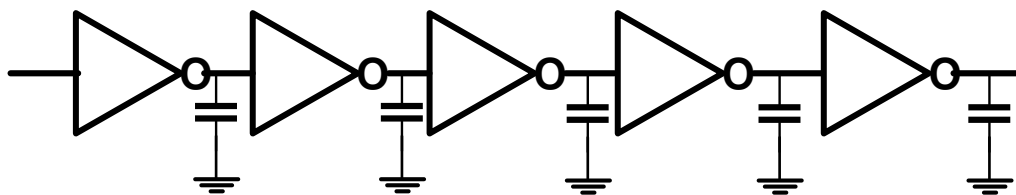


Fig. 5. Inverter Chain Generating Delay

Observation and Results

Hard vs. Soft Switching:

When insufficient dead time is used, the converter operates under hard switching, characterized by significant overlap of drain current and switch-node voltage. With properly designed dead time, soft switching is achieved, where the switch-node voltage transitions prior to device turn-on. This significantly reduces switching losses and device stress.

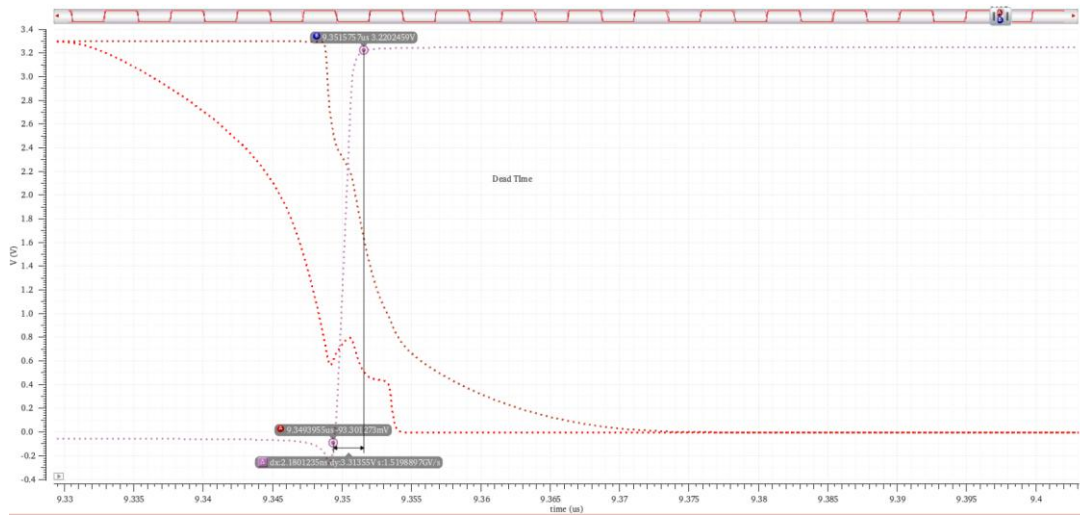


Fig. 6. Hard Switching

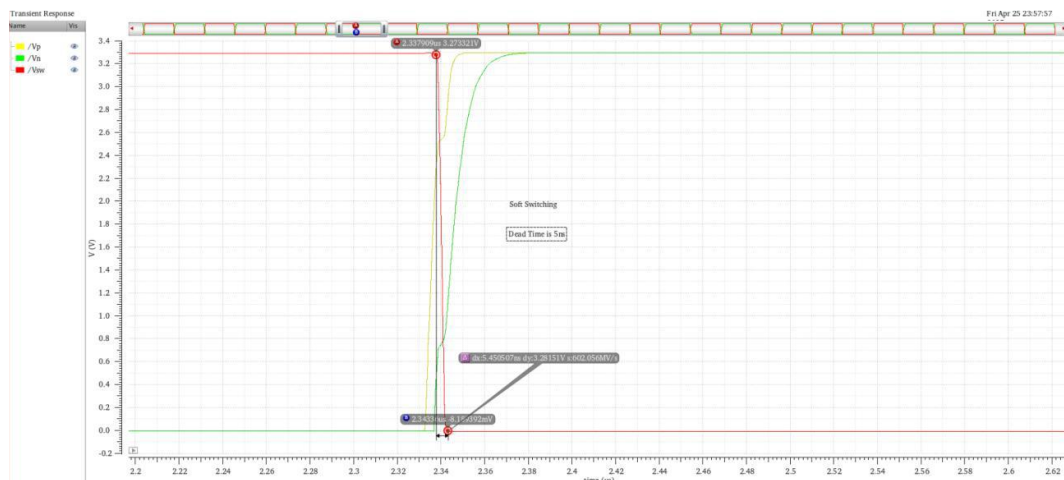


Fig. 7. Soft Switching

Dead time is 2ns and Time period is 666ns. Dead time percentage = $5.45/666 * 100 = 0.8183\%$

Gate Capacitance and Resistance of PMOS and NMOS Drivers:

PMOS M1:

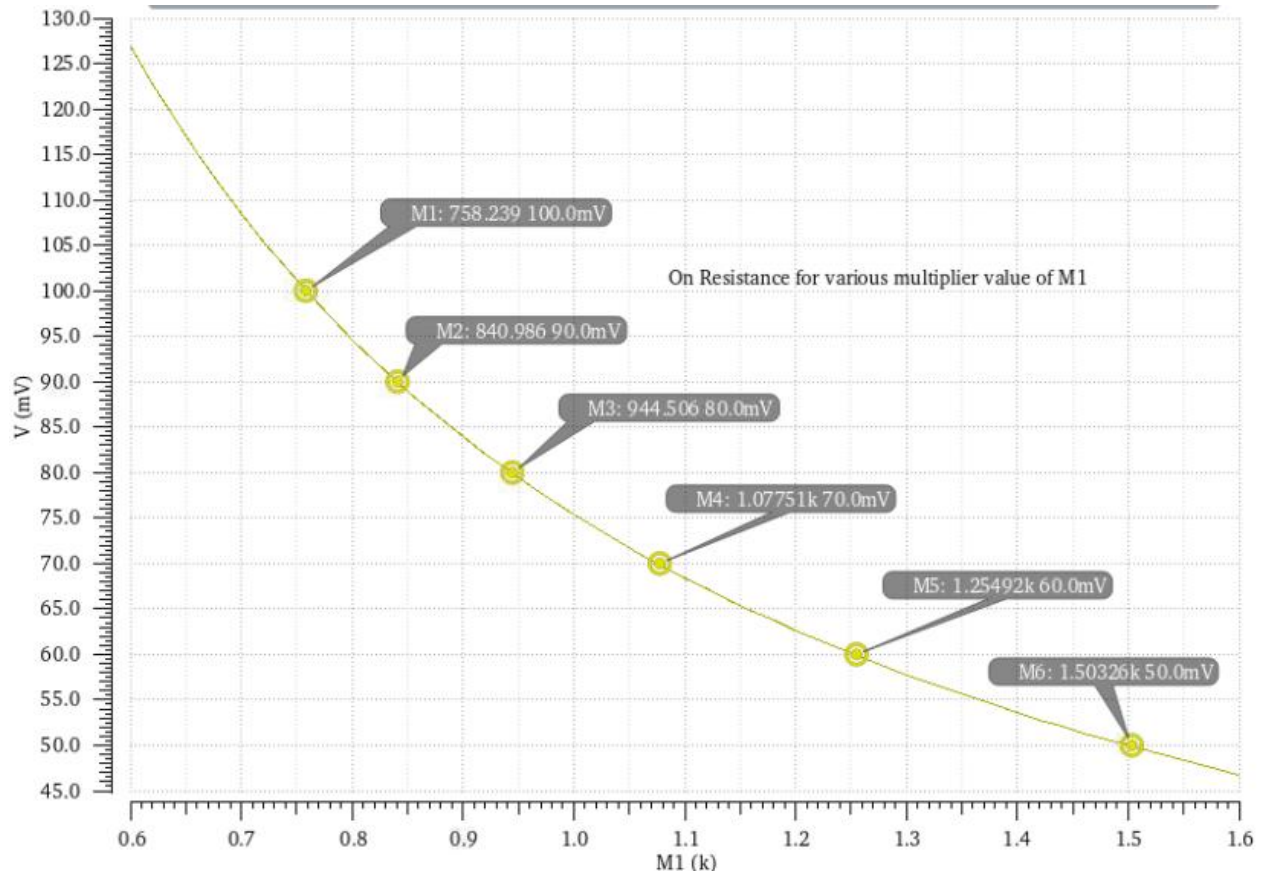


Fig. 8. ON Resistance (mV/A) for various multiplier values of MOSFET M1 PMOS

$R_{on,M1}$	$C_{gs,M1}$	$C_{gd,M1}$	$C_{ds,M1}$	Multiplier of M1
100m Ω	55.2pF	51.84pF	109.2fF	758
90m Ω	61.17pF	57.6pF	114.1fF	841
80m Ω	68.64pF	64.82pF	121.5fF	945
70m Ω	78.2pF	74.05pF	132.2fF	1078
60m Ω	90.92pF	86.33pF	147.8fF	1255
50m Ω	108.9pF	103.7pF	171.7fF	1503

NMOS M2:

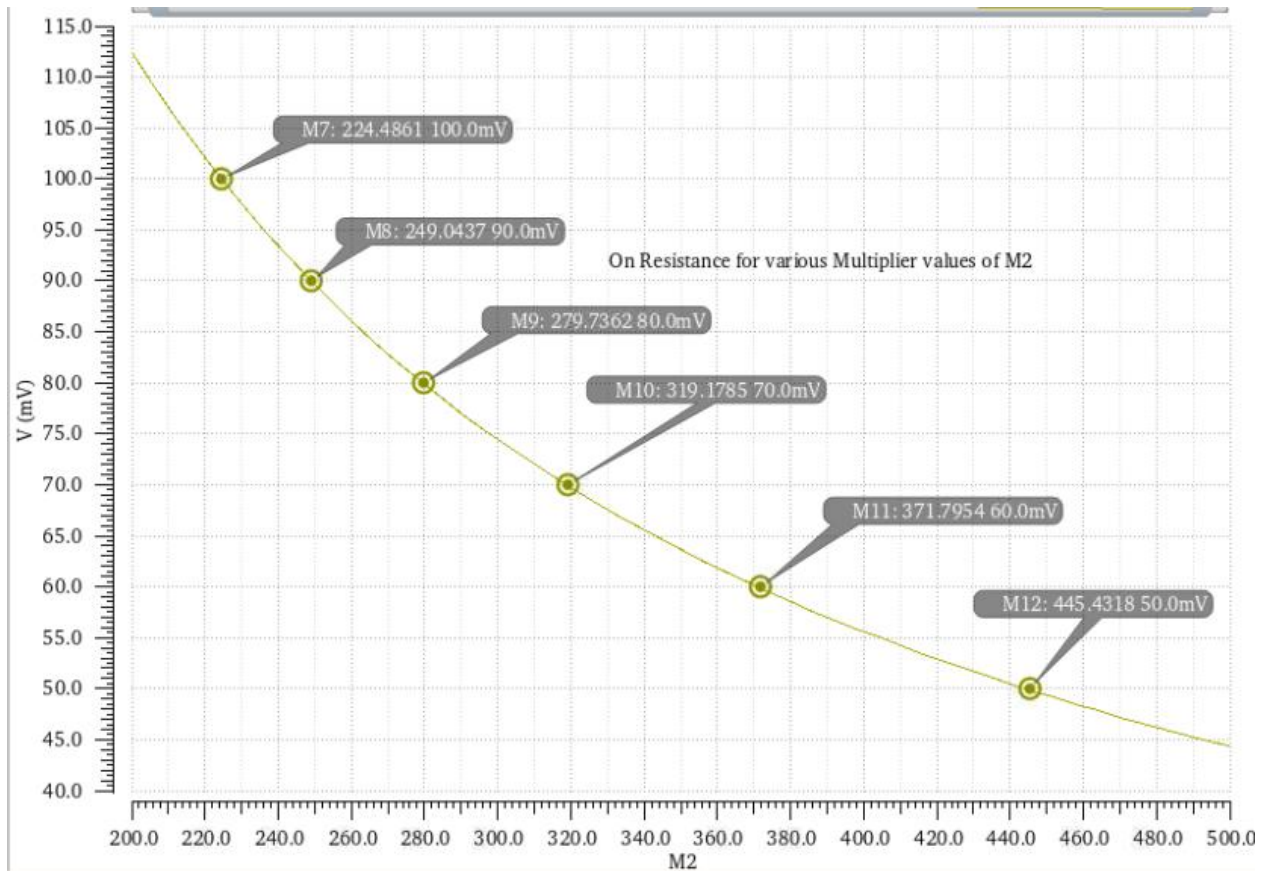


Fig. 9. ON Resistance (mV/A) for various multiplier values of MOSFET M2 NMOS

$R_{on,M2}$	$C_{gs,M2}$	$C_{gd,M2}$	$C_{ds,M2}$	Multiplier of M2
100m Ω	17.62pF	16.46pF	372.4fF	224
90m Ω	19.57pF	18.31pF	417.4fF	249
80m Ω	22pF	20.6pF	473.1fF	280
70m Ω	25.05pF	23.48pF	543.4fF	319
60m Ω	29.2pF	27.39pF	638.8fF	372
50m Ω	34.91pF	32.79pF	770.4fF	445

Sizing of MOSFETs:

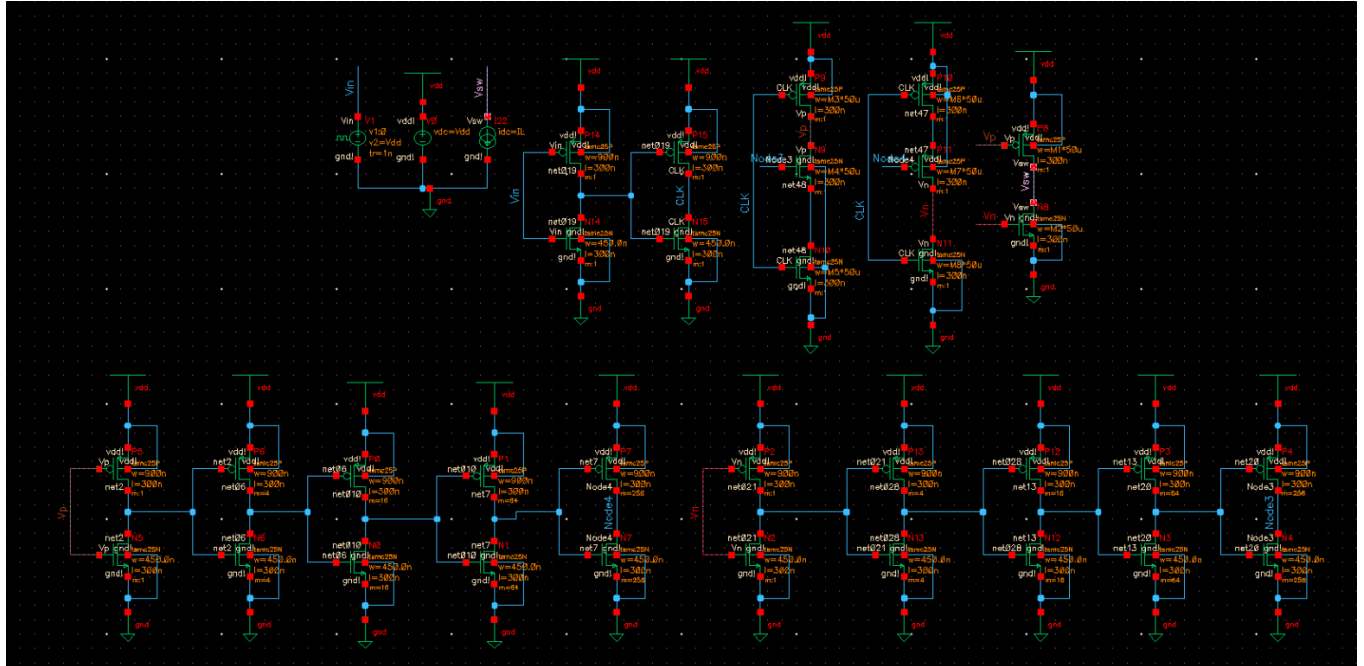


Fig. 10. Driver Schematics of MOSFET with sizes

MOSFET	Multiplier (with 10 um size of W)
M3	667
M4	230
M5	230
M6	200
M7	200
M8	70

Linearized DC Model with and without PWM:

The linearized small-signal model of the power stage exhibits a second-order LC response with a resonant peak determined by the inductor, capacitor, and ESR. Incorporating the PWM modulator scales the overall loop gain by the inverse of the ramp amplitude while leaving the phase response largely unchanged, confirming theoretical expectations.

Without PWM Modulator:

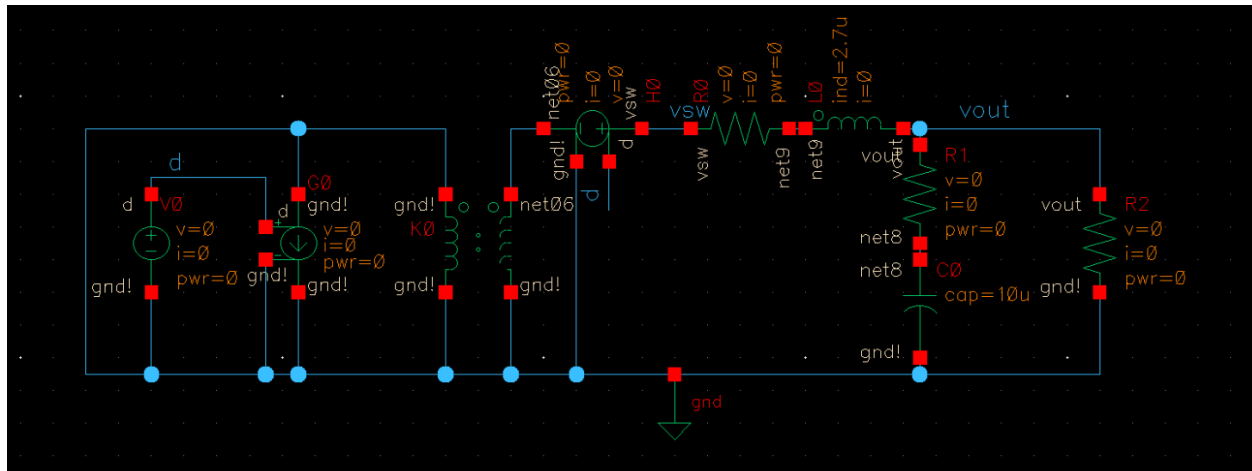


Fig. 11. Schematics of Linearized AC Model without PWM Modulator

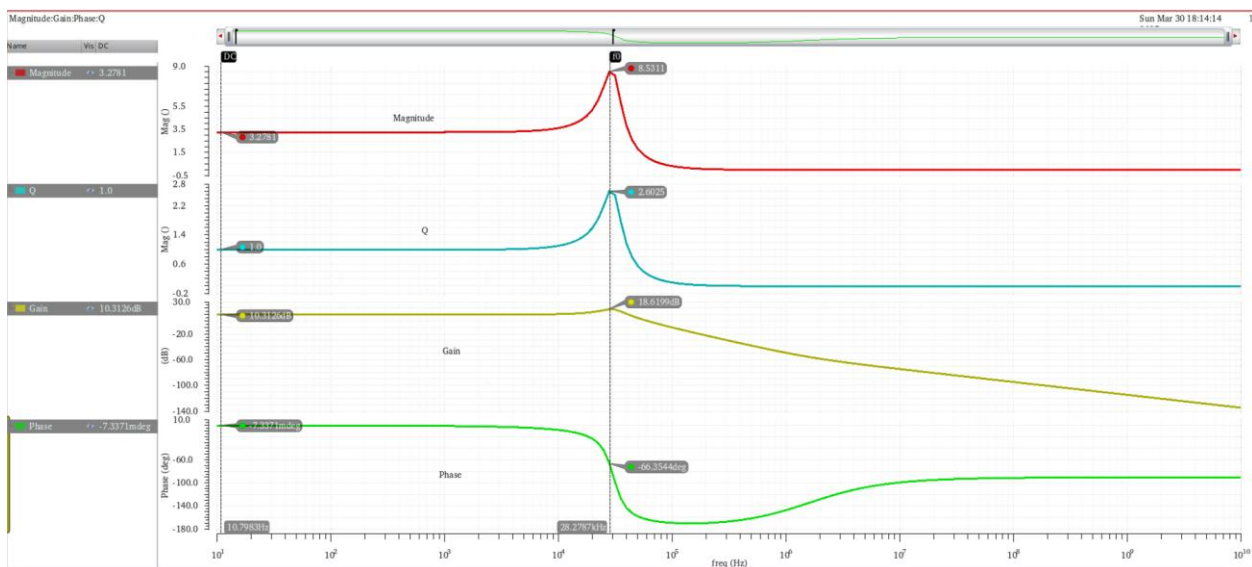


Fig. 12. AC Plot of Linearized AC Model without PWM Modulator

Observed Magnitude at $f_0 = 8.5311\text{V}$

Expected Magnitude at $f_0 = 8.5988\text{V}$

Observed Q = 2.6025

Expected Q value is 2.606

Observed $f_0 = 28.2787\text{KHz}$

Expected $f_0 = 30.67\text{KHz}$

With PWM Modulator:

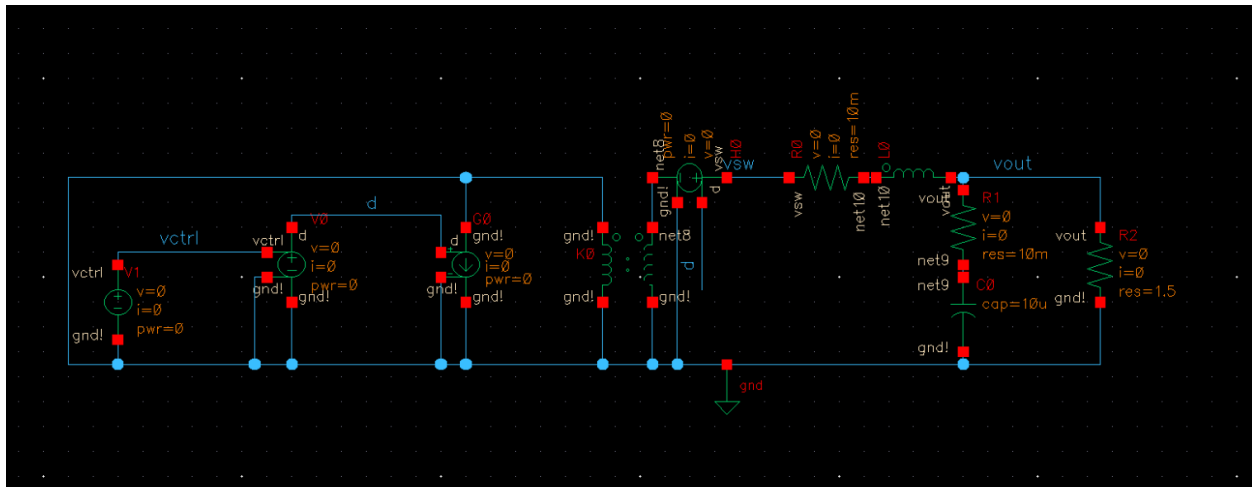


Fig. 13. Schematics of Linearized AC Model with PWM Modulator

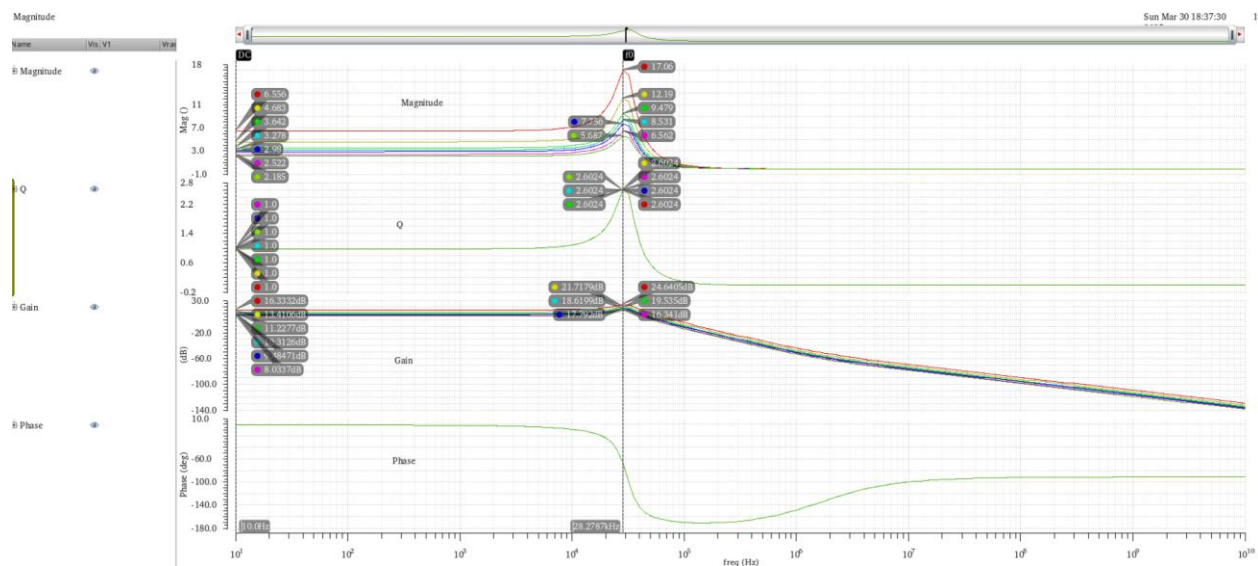


Fig. 14. AC Plot of Linearized AC Model with PWM Modulator with V_{ramp} ranging from 0.5V – 1.5V

PWM Modulation voltage V_{ramp} is a constant component with respect to frequency getting divided to the output voltage, varying the voltage component of PWM doesn't vary the results in no difference in Q and Phase values, but the magnitude and Gain varies linearly with change in V_{ramp} of PWM Modulation.

Compensation Network with Phase Margin $\approx 0^\circ$ and 60° :

With a phase margin near 0° , the closed-loop response shows pronounced ringing and long settling times. Increasing the phase margin to approximately 60° results in a well-damped response with reduced overshoot and faster settling, demonstrating the importance of proper compensation design.

Low Phase Margin:

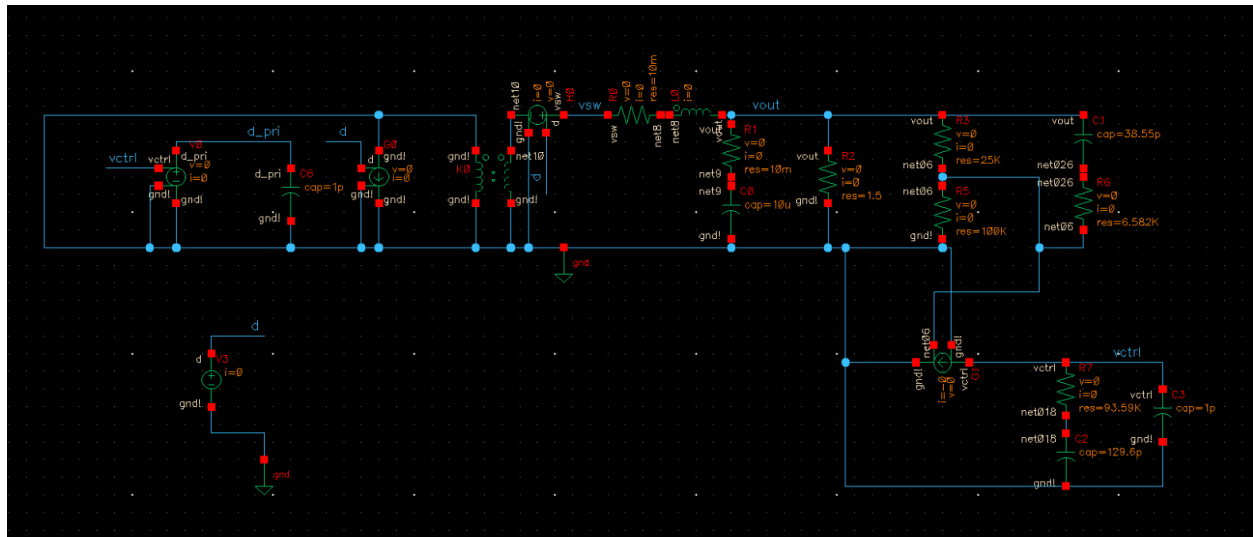


Fig. 15. Schematics of Voltage Mode Buck Converter with Low PM

Parameters	Values
C2	1pF
Ci	129.6pF
Cd	46.32pF
Rp	93.57kOhms
R1	25kOhms
R2	100kOhms
Rd	6.58kOhms
gm	86.47uS

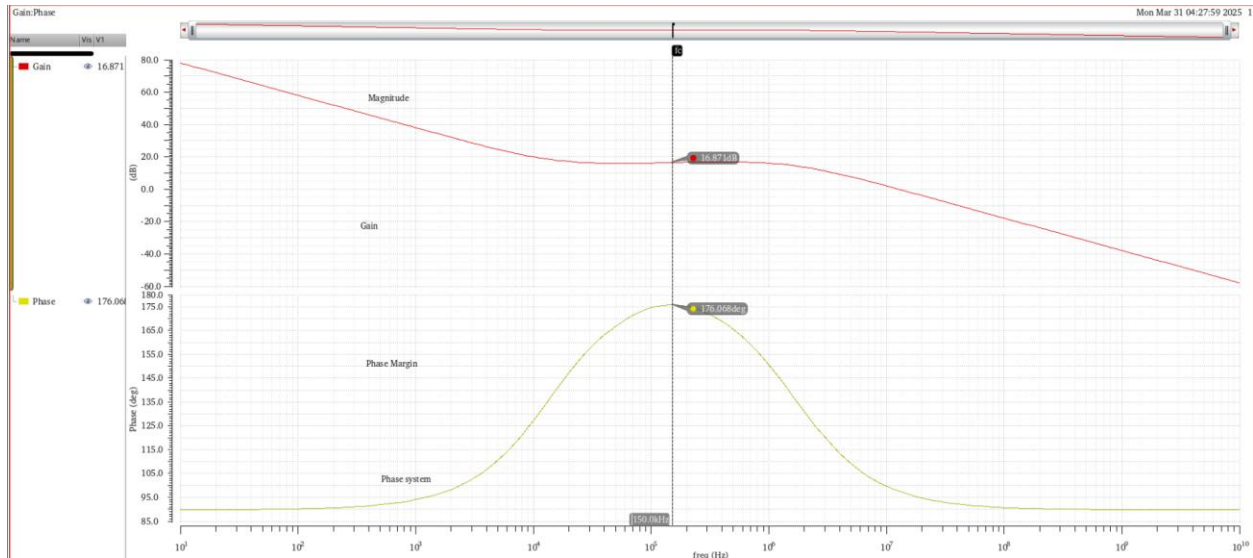


Fig. 16. AC Plot of Compensation Network with Low PM

Magnitude Gain at 150KHz is 7 in compensation network

Gain in dB = $20\log(7) = 16.9\text{dB}$

Observed Gain at $f_c = 16.871\text{dB}$

Observed Phase at $f_c = 176$ degrees

Expected Phase at $f_c = 5 + 169.96 = 174.96$ degrees

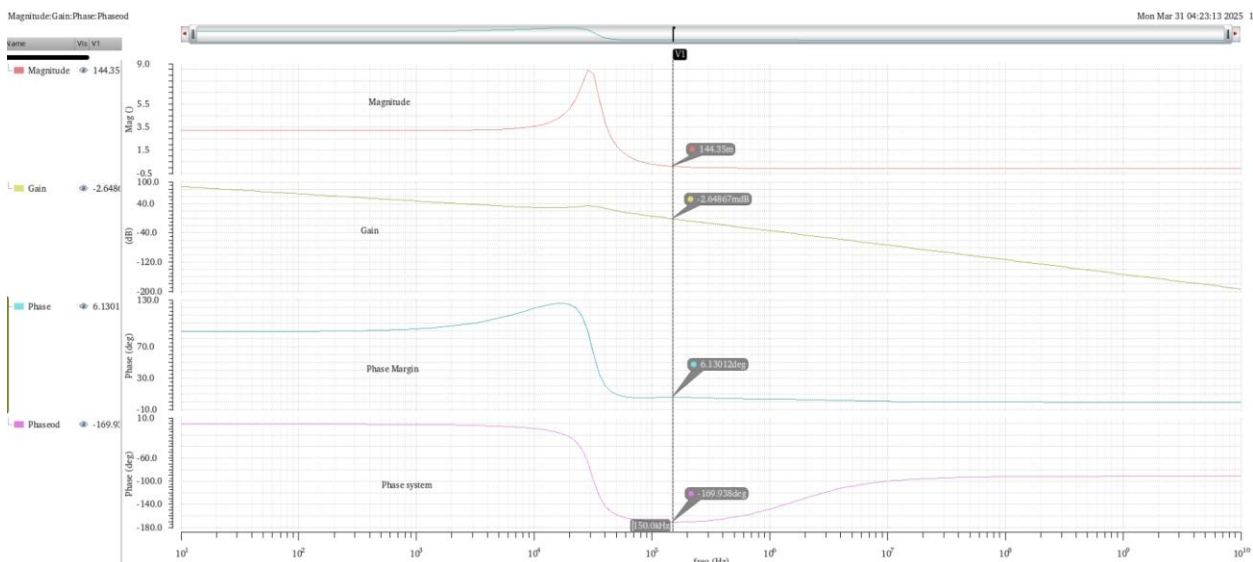


Fig. 17. AC Plot of Whole Buck Converter System with Low PM

Unity Gain Bandwidth of whole system is 150kHz

Observed Phase Margin = 6.13 degrees

Expected Phase Margin = 5 degrees

Observed System Phase at $f_c = 169.938$ degrees

Expected System Phase at $f_c = 169.96$ degrees

High Phase Margin:

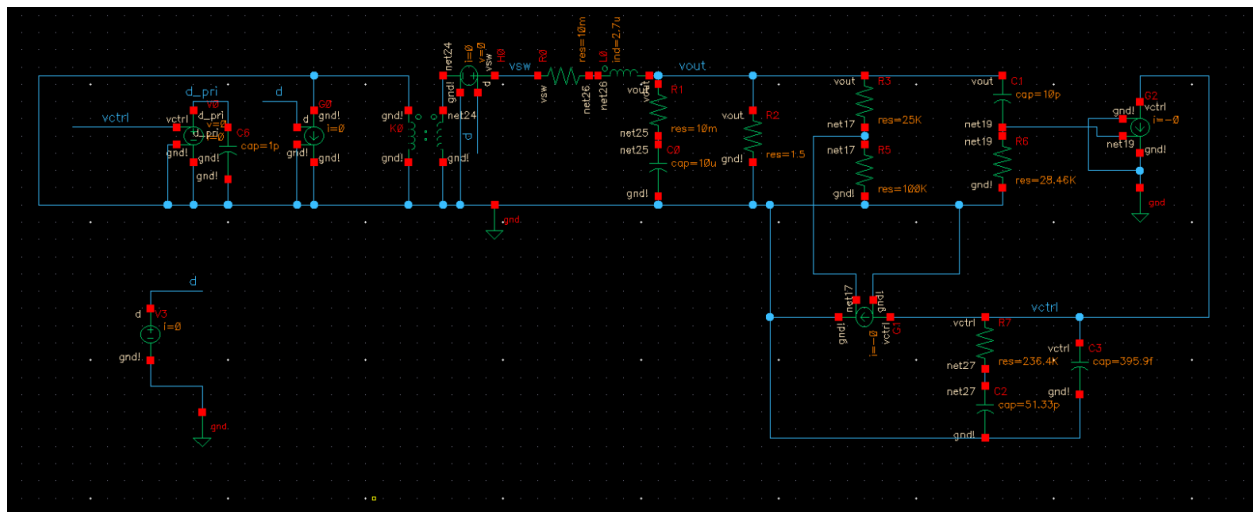


Fig. 18. Schematics of Voltage Mode Buck Converter with High PM

Parameters	Values
C2	395.7fF
Ci	51.3pF
Cd	10pF
Rp	236.39kOhms
R1	25kOhms
R2	100kOhms
Rd	28.45kOhms
gm1	10uS
gm2	103.3uS

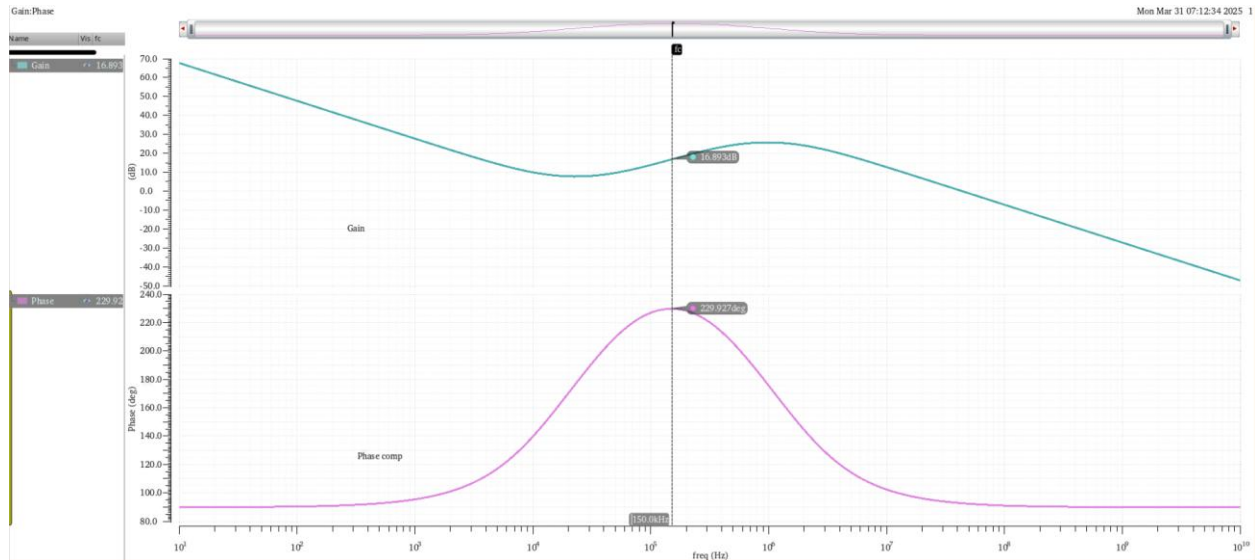


Fig. 19. AC Plot of Compensation Network with High PM

Magnitude Gain at 150KHz is 7 in compensation network

Gain in dB = $20\log(7) = 16.9\text{dB}$

Observed Gain at $f_c = 16.893\text{dB}$

Observed Phase at $f_c = 229.927$ degrees

Expected Phase at $f_c = 60 + 169.96 = 229.96$ degrees

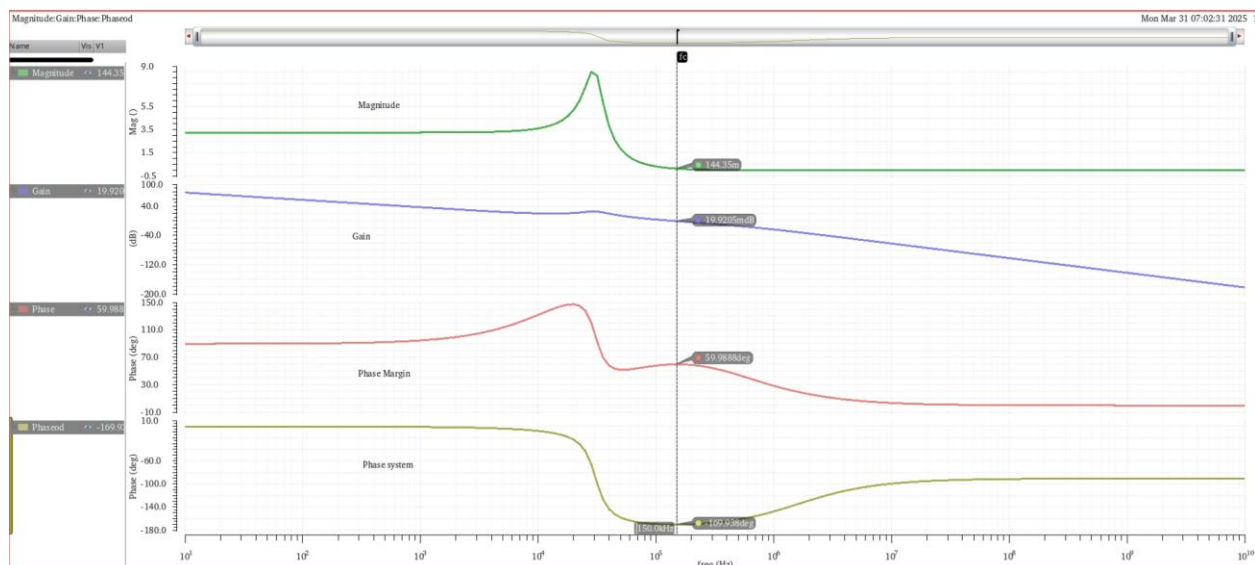


Fig. 20. AC Plot of Whole Buck Converter System with High PM

Unity Gain Bandwidth of whole system is 150kHz

Observed Phase Margin = 59.988 degrees

Expected Phase Margin = 60 degrees

Observed System Phase at f_c = 169.938 degrees

Expected System Phase at f_c = 169.96 degrees

Switching Observations at Maximum and Minimum Currents:

Switch-node voltage and gate-drive waveforms confirm correct sequencing of the high-side and low-side devices. The measured waveforms validate dead-time insertion and soft-switching operation at both minimum and maximum load currents.

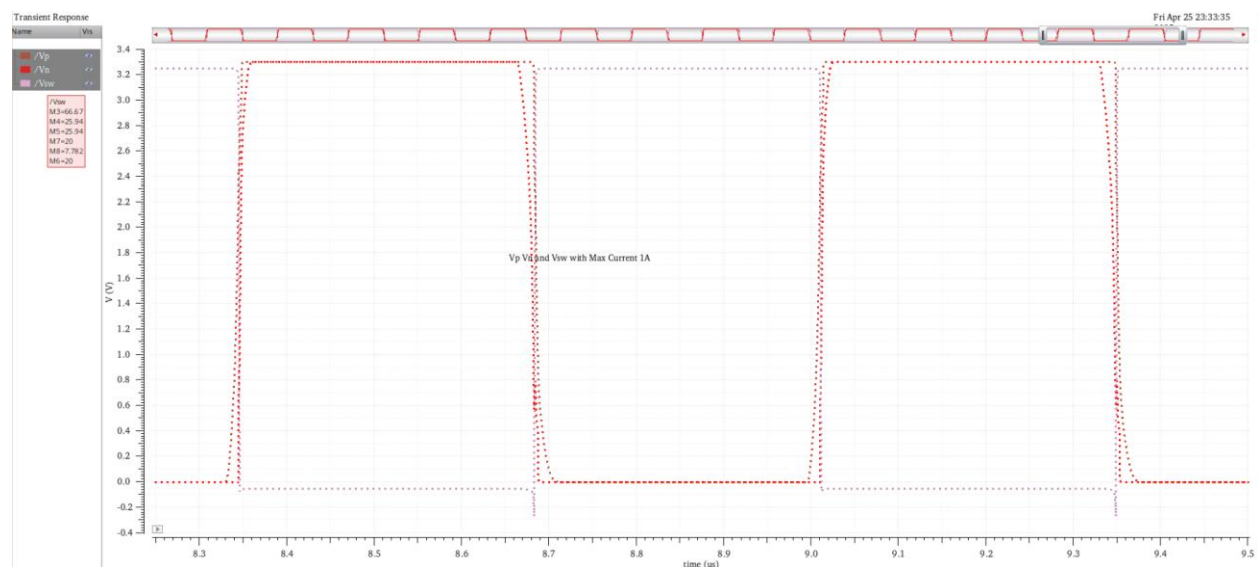


Fig. 21. Switching Voltage at Maximum Current

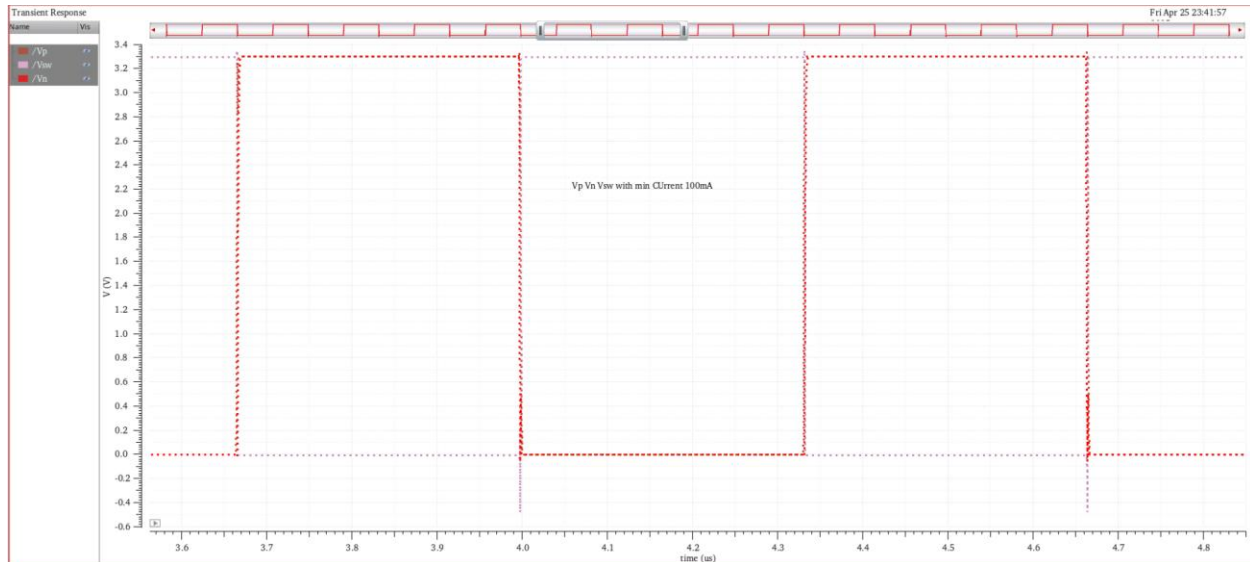


Fig. 22. Switching Voltage at Minimum Current

Load Observations:

The output voltage ripple and inductor current ripple closely match analytical predictions based on component selection. This agreement verifies the correctness of the power-stage design and small-signal assumptions.

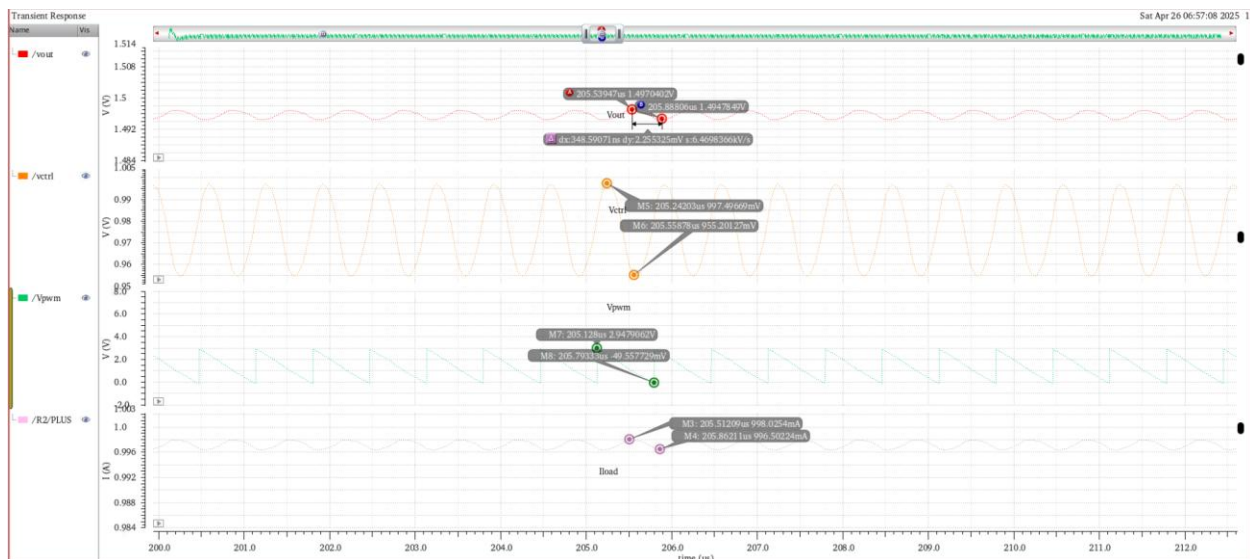


Fig. 23. Ripple Voltage Observation at Load

$$\Delta I_L = \frac{V_{in}(1-D)D}{L} T_{sw} = 201.66mA$$

$$\Delta V_{out} = \frac{\Delta I_L}{C * 8 * F_{sw}} = 1.68mV$$

	Calculated values	Values from simulation
Vout ripple	1.68mV	1.1275mV
Inductor current ripple	201.66mA	1.497mA

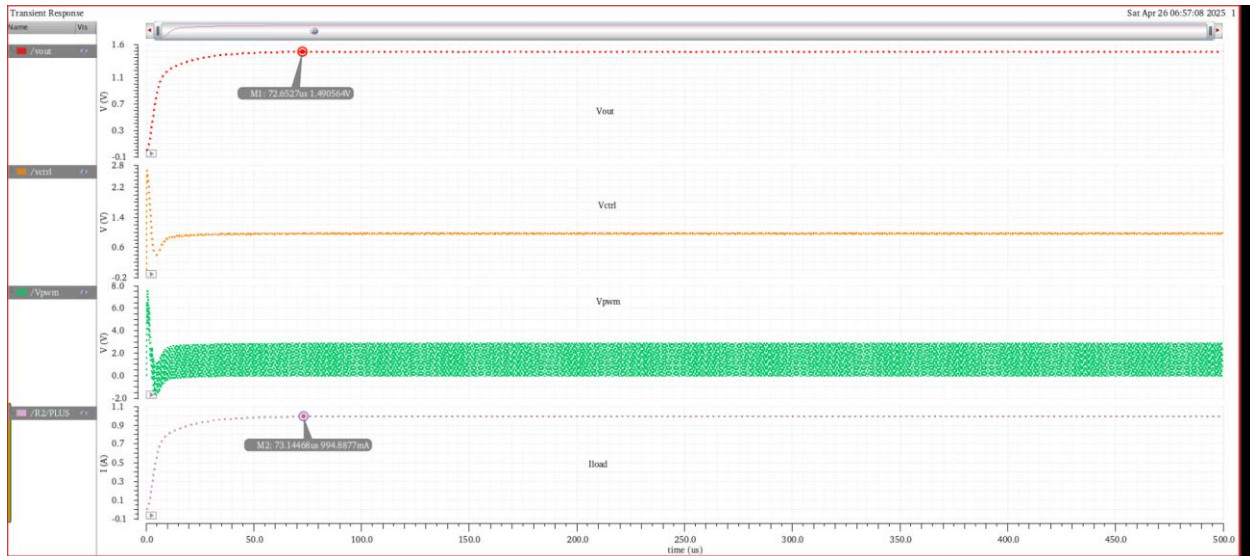


Fig. 24. Plot of Vout, Vctrl, Vpwm and Iload

Efficiency and Losses:

Efficiency is highest at moderate load currents, where switching and conduction losses are balanced. At light load, switching and gate-drive losses dominate, while at heavy load, conduction losses become significant. The observed efficiency trends align with established buck-converter theory.

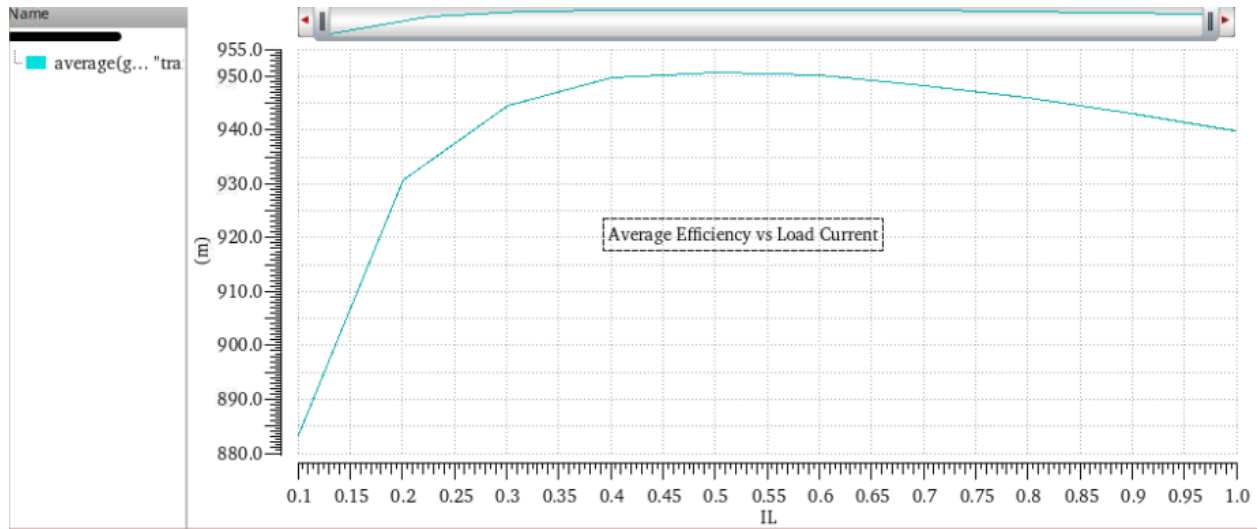


Fig. 25. Plot of Efficiency vs Load Current

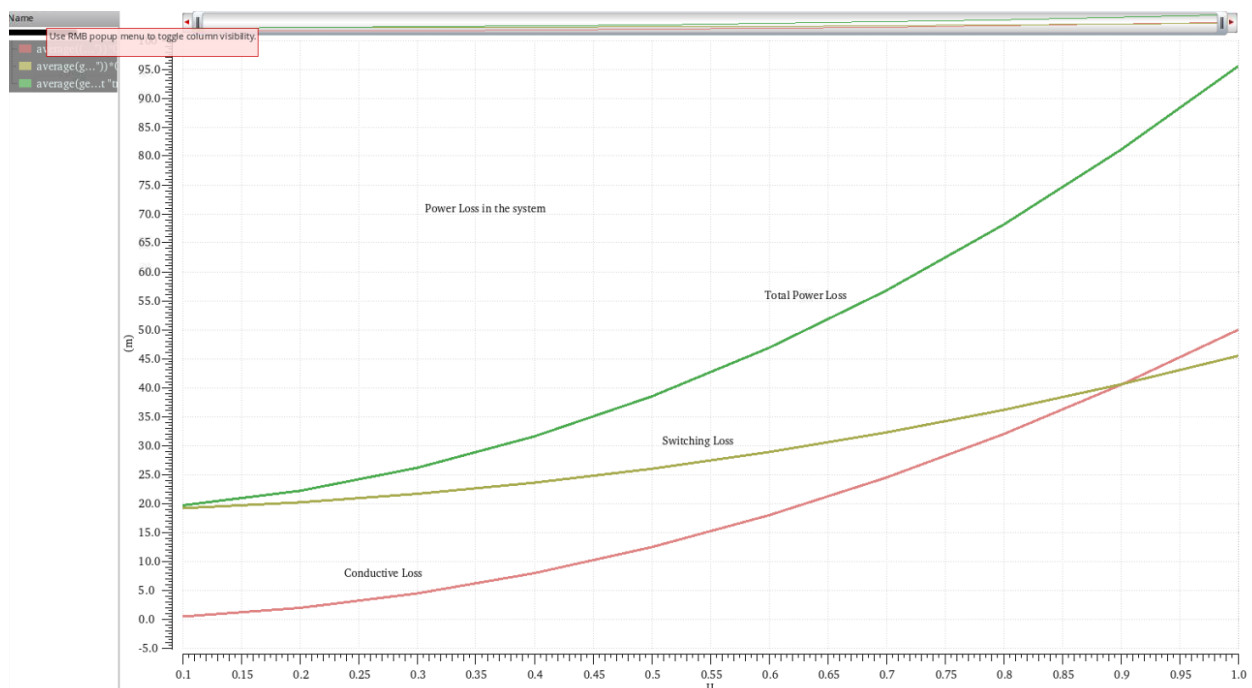


Fig. 26. Plot of Power Loss in the system vs Load Current

Conductive Loss increases with Load Current as it is quadratically dependent, with increase in current the loss also increases but not quadratically because of switching loss which is linearly or sub linearly dependent on load current due to larger deadtime at lower currents which increase the loss when load current is low.

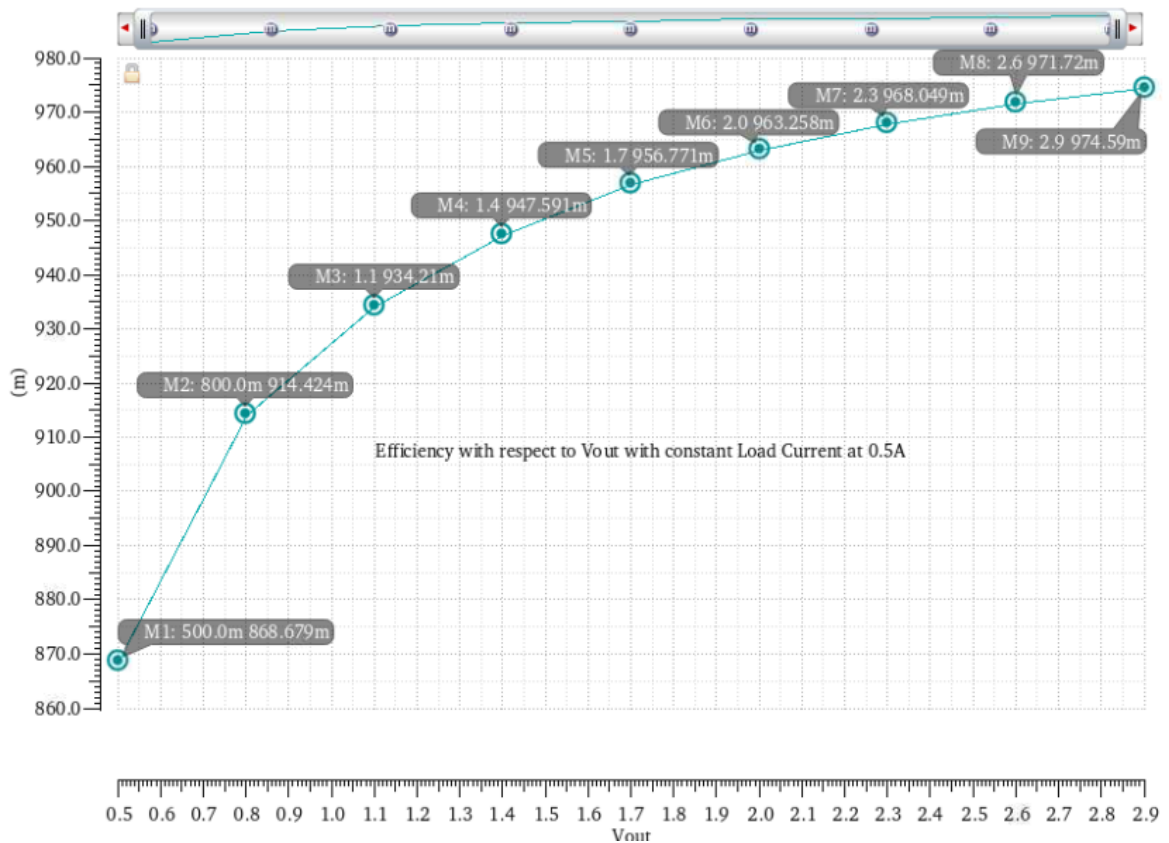


Fig. 27. Efficiency vs Vout at Mid Load of 0.5A

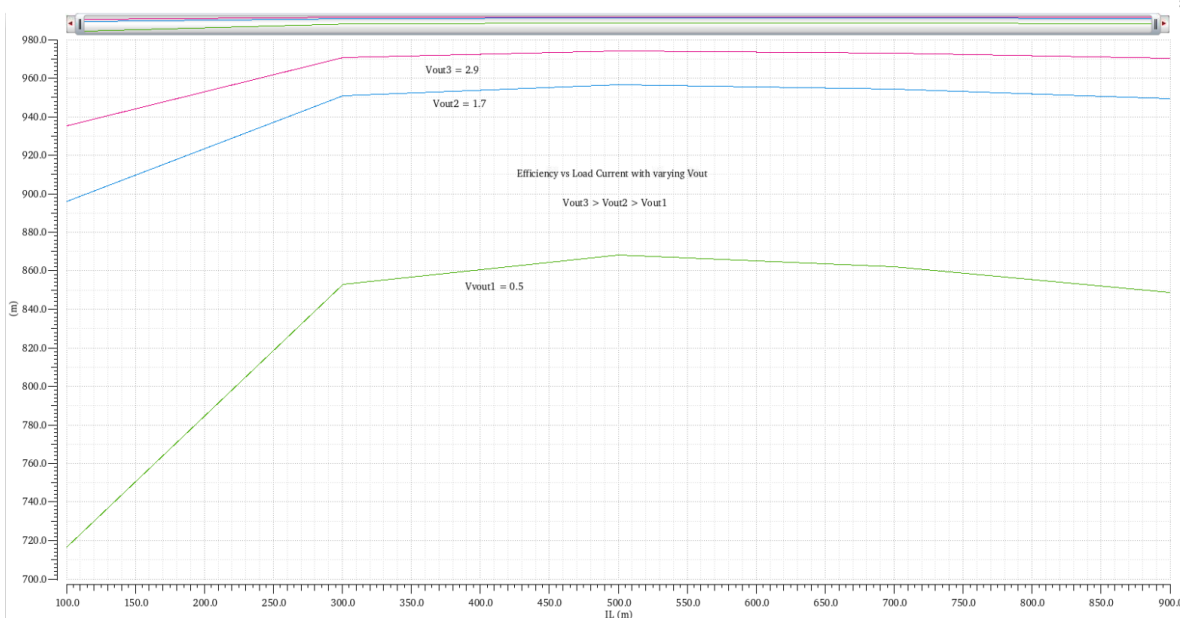


Fig. 28. Efficiency vs Vout at Low, Mid and High Load

Transient Analysis Observations:

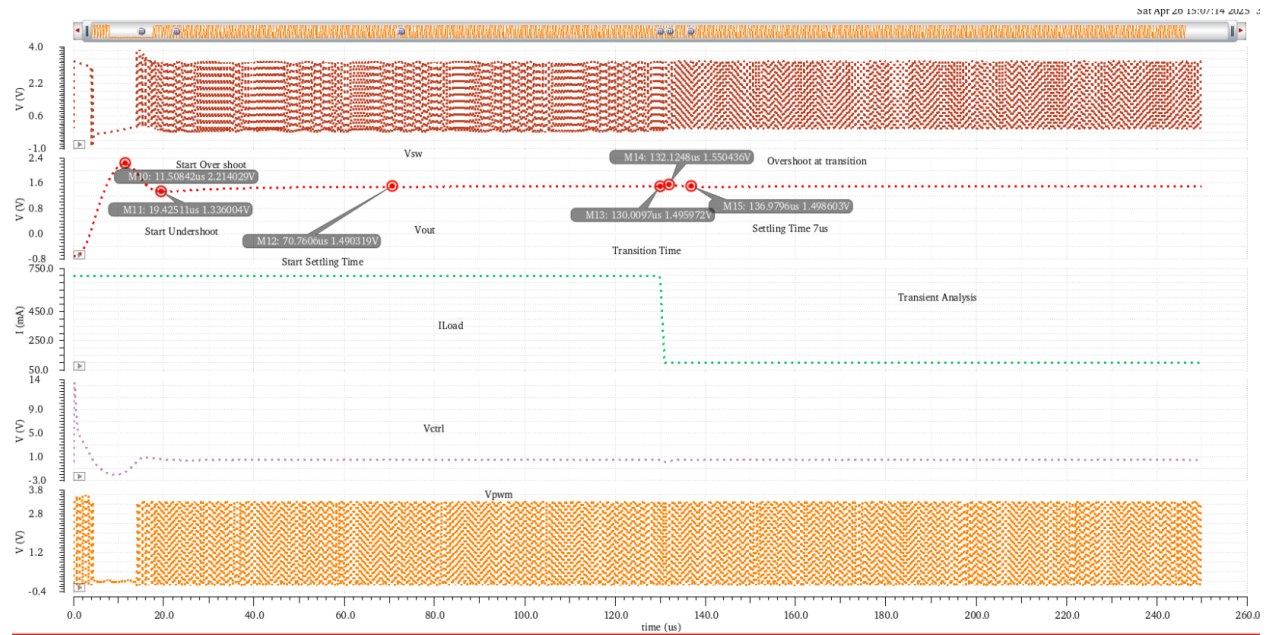


Fig. 29. Transient Analysis of Switching Voltage, Output Voltage and Load Current

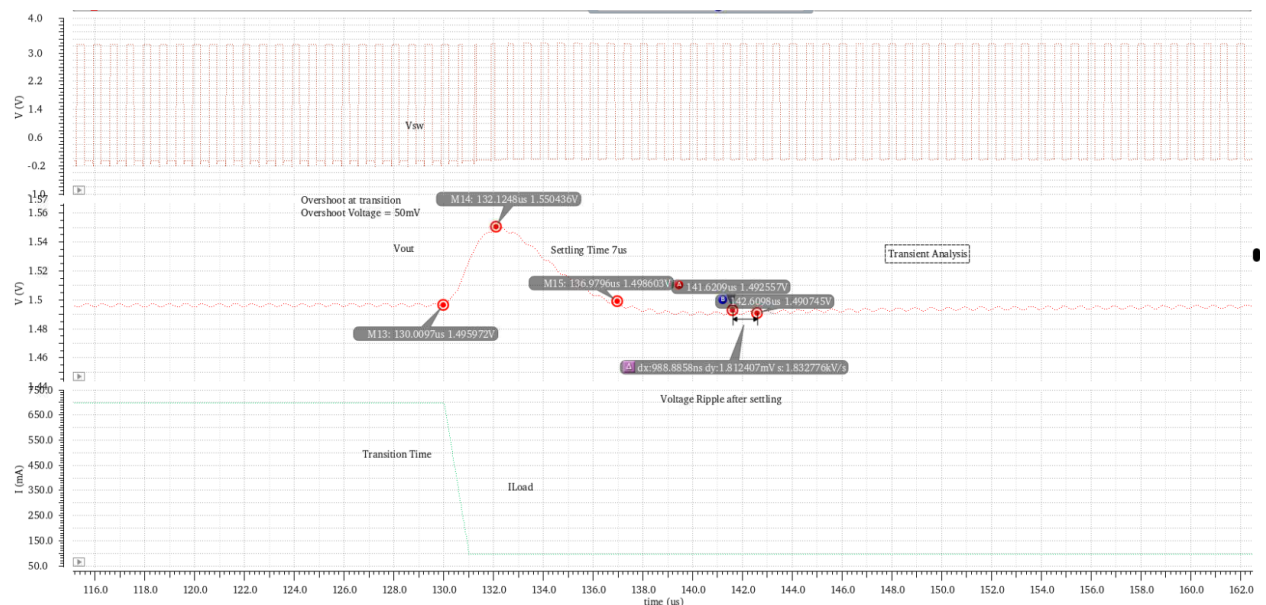


Fig. 30. Overshoot and Undershoot Markings Zoomed in

Voltage Ripple of $1.8\text{mV} < 1\%$ of V_{out}

Settling Time = $6.97\mu\text{s}$

Overshoot Voltage = 50mV

Learning Outcomes

This work provided a comprehensive understanding of the end-to-end design process of a high-frequency buck converter suitable for PMIC applications. Key learning outcomes include:

- **System-level design thinking:** The project reinforced how electrical specifications such as ripple limits, bandwidth, and efficiency targets directly influence component selection, switching frequency, and control architecture.
- **Small-signal modeling and stability analysis:** By deriving and analyzing the linearized power-stage model, the relationship between LC resonance, ESR, quality factor, and loop stability became clear. This enabled informed compensation design rather than trial-and-error tuning.
- **Compensation network design:** Designing Type-III compensation networks highlighted the practical trade-offs between phase margin, bandwidth, noise attenuation, and transient response. The comparison between low and high phase margin cases clearly demonstrated how stability margins affect ringing, overshoot, and settling time.
- **Gate-driver and switching behavior:** The impact of MOSFET gate charge, dead time, and driver strength on switching losses and waveform integrity was studied in detail. The distinction between hard and soft switching reinforced the importance of timing control in high-frequency converters.
- **Efficiency analysis:** Evaluating efficiency across load conditions provided insight into how conduction losses dominate at high load, while switching and gate-drive losses dominate at light load. This understanding is essential for designing converters optimized for real-world operating profiles.
- **Integration of analog and digital blocks:** The interaction between analog control loops and digital timing circuitry emphasized the mixed-signal nature of PMIC design and the need for careful coordination between control, switching, and logic blocks.

Overall, the project strengthened both theoretical understanding and practical intuition for designing stable, efficient switching power converters.

Conclusion and Future Scope

Metric	Target	Achieved	Status
Input Voltage (V_{in})	3.3 V	3.3 V	Met
Output Voltage (V_{out})	1.5V	1.5V	Met
Load Current Range	0.1A – 1A	0.1A – 1A	Met
Switching Frequency	1.5MHz	1.5MHz	Met
Inductor Current Ripple	<200mA	1.497mA	Met
Output Voltage Ripple	<1.68mV	1.125mV	Met
Control Bandwidth (U_{GB})	150KHz	150KHz	Met
Phase Margin (Low-PM Case)	5°	6.13°	Met
Phase Margin (High-PM Case)	60°	59.988°	Met
Switching Behavior	Soft Switching	Soft Switching	Met
Shoot-Through	None	None	
Efficiency	Acceptable Trend	Trend Observed	Met

A complete voltage-mode buck converter was designed and evaluated starting from fundamental specifications and progressing through modeling, control design, and switching analysis. The converter achieves stable closed-loop regulation, controlled switching behavior through dead-time management, and high efficiency across a wide range of load currents.

The results demonstrate that proper small-signal modeling and compensation are critical for achieving predictable stability and transient performance, while careful gate-driver design is essential for minimizing switching losses and device stress. The close agreement between analytical predictions and simulation results validates the design methodology employed in this work.

This design framework is directly applicable to integrated PMIC development and can be extended in several directions. Future work may include current-mode or peak-current control, adaptive dead-time or PWM/PFM operation for improved light-load efficiency, and layout-aware parasitic modeling to account for on-chip interconnect effects. These extensions would further bridge the gap between conceptual design and silicon implementation in advanced power management systems.